

Flow-PPO Experiment Analysis

Why Every Run Collapses, and What to Fix First

Flow-Factory

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Abstract

Source: W&B project Flow-PPO (entity 315229706-xi-an-jiaotong-university-), 15 runs across 6 commits, SD3.5-Medium LoRA + PickScore. Data was pulled and plotted with the repository `.venv`; raw dumps live under `.scratch/ppo_wandb/`. This note diagnoses a failure mode common to *every* configuration tried so far—a fast rise to a reward ceiling followed by a slow collapse—identifies its mechanism (reward over-optimization with no KL anchor, plus a critic that barely beats a mean baseline), and proposes a phased experiment plan that fixes most variables and solves the collapse first.

TL;DR

1. **Every configuration shows the same curve:** a fast rise to an eval-PickScore ceiling of ~ 0.83 – 0.84 (from the 0.769 base), then a slow monotone collapse. No config sustains gains; the longer a run trains, the more it gives back (Fig. 1, Fig. 2).
2. **The collapse is reward over-optimization (reward hacking / distribution drift), not numerical instability.** The PPO ratio sits at ~ 1.0 and policy grad-norm at $\sim 3e-4$ throughout—no blow-up, just a tiny, persistent, *uncorrected* push toward the reward model. Crucially, `kl_beta = 0` in *every* run: the one regularizer that prevents this was never enabled (Fig. 3).
3. **The value critic barely earns its keep.** For the standalone critic, explained variance ≈ 0 —it just predicts the running mean reward, so PPO degenerates into REINFORCE-with-a-mean-baseline. The **backbone critic (Scheme B)** is measurably better (explained variance $\rightarrow 0.2$); the **single-SDE bandit critic** is worse (negative explained variance) (Fig. 4).
4. **Recommendation:** stop sweeping warmup / re-warmup / critic-follow knobs. Fix everything to one simple baseline and solve the collapse first with a KL anchor + an early-stop metric. Only then optimize the critic body. See §6.

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1 What was run (config matrix)

All runs are SD3.5-Medium, LoRA, PickScore reward, 10 inference steps (except two 40-step backbone runs), `group_size` 1. The axes that actually vary:

- **Warmup:** `critic_warmup_steps` (warm); periodic re-warmup `critic_warmup_interval` (rwI) / `critic_rewarmup_steps` (rwS).
- **Critic-follows-policy:** `update_critic_in_optimize` (updC) — True = critic trained jointly each optimize step; False = critic only via warmup/re-warmup bursts.
- **Value-net structure:** `critic_type` — `standalone` (lightweight PMA net) vs `backbone` (Scheme B: diffusion backbone + value head, the “Explicit Critic” direction) vs `backbone_branches`.
- **Algorithm:** `sde_step_selection` — `global` (full-SDE PPO) vs `per_sample` (single-step SDE / bandit).
- Older 4bdec9d2 runs predate the `critic_type` / `sde_step_selection` fields; they are the implicit standalone + global path.

Table 1: Configuration matrix (sorted by run length). “—” = field absent at that commit.

run	commit	critic	warm	rwI	rwS	updC	sde_sel	infer	kl_beta	gscale
rewarmup-10-6	87dc126c	standalone	30	10	6	False	global	10	0	1
default-cfg	87dc126c	standalone	30	0	0	True	global	10	0	4.5
rewarmup-10-6_update-critic	87dc126c	standalone	30	10	6	True	global	10	0	1
sd3-5_..._512_q1_warm50	4bdec9d2	standalone	50	—	—	—	global	10	0	1
default-standalone	9c54d8f0	standalone	30	0	0	True	global	10	0	1
sd3-5_..._256_q1	4bdec9d2	standalone	30	—	—	—	global	10	0	1
backbone (×2)	d0482331	backbone	30	0	0	True	global	10	0	1
sd3-5_..._512_q1	4bdec9d2	standalone	30	—	—	—	global	10	0	1
single-sde	9b032788	standalone	30	0	0	True	per_sample	10	0	1
sd3-5_..._rewarmup-100-10	9b032788	standalone	30	100	10	False	global	10	0	1
rewarmup-10-5	87dc126c	standalone	30	10	5	False	global	10	0	1
sd3-5_..._512_q4	4bdec9d2	standalone	30	—	—	—	global	10	0	1
explicit_critic	81511ef7	backbone_branches	15	0	0	True	global	40	0	1
backbone_infer40	81511ef7	backbone	30	0	0	True	global	40	0	1

Commit legend. 4bdec9d2 initial critic PPO (value net + GAE); 9b032788 `group_size`=1, `samples`=128; 87dc126c project rename (after multi-axis logging); d0482331 add backbone critic (Scheme B); 9c54d8f0 backbone-critic LoRA align; 81511ef7 dedup / collapse PPO critic-mode branches.

The design-space explosion is real. The axes multiply out to warmup {30, 50, 15}, re-warmup {off, 10/5, 10/6, 100/10}, updC {T, F}, critic {standalone, backbone, branches}, algo {global, per_sample}, and infer {10, 40}. With 15 runs this grid is far from covered, and **the one variable theory says matters most (kl_beta) is fixed at 0 everywhere.**

2 Finding 1 — Universal rise-then-collapse

The two long runs make the trajectory clear:

- default-cfg (2728 steps): eval peaks **0.838 @ step 290**, decays to **0.792**.

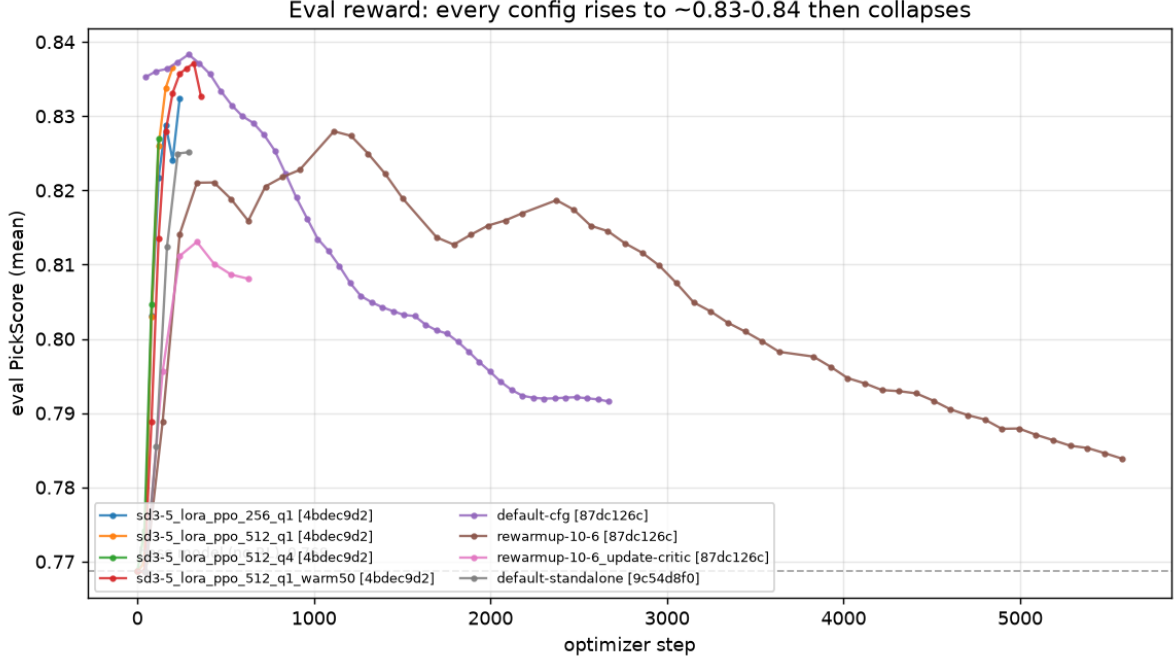


Figure 1: Eval PickScore vs. optimizer step. Every run rises from the 0.769 base to ~ 0.83 –0.84, then declines. The two long runs (**default-cfg** purple, **rewarmup-10-6** brown) make the trajectory unambiguous.

- **rewarmup-10-6** (5611 steps, longest): eval peaks **0.828 @ step 1113**, decays to **0.784**; train reward collapses hardest, $0.814 \rightarrow 0.650$.

Short runs (< 400 steps) look “fine” only because they were stopped near the peak; their train reward is already turning over. There is no qualitative difference between configs—only “how long until it rolls over.”

Table 2: Peak vs. final reward (sorted by eval peak). Base model = 0.769.

run	commit	max_step	eval_peak	eval_final	train_peak	train_final
default-cfg	87dc126c	2728	0.838	0.792	0.836	0.752
sd3-5_..._512_q1_warm50	4bdec9d2	398	0.837	0.833	0.836	0.762
sd3-5_..._512_q1	4bdec9d2	209	0.837	0.837	0.831	0.819
sd3-5_..._256_q1	4bdec9d2	243	0.832	0.832	0.830	0.828
rewarmup-10-6	87dc126c	5611	0.828	0.784	0.814	0.650
sd3-5_..._512_q4	4bdec9d2	131	0.827	0.827	0.836	0.808
default-standalone	9c54d8f0	350	0.825	0.825	0.827	0.782
rewarmup-10-6_update-critic	87dc126c	706	0.813	0.808	0.816	0.767
backbone	d0482331	227	—	—	0.825	0.808
single-sde	9b032788	188	—	—	0.777	0.741
explicit_critic	81511ef7	106	—	—	0.786	0.777

3 Finding 2 — The collapse is over-optimization, not instability

Windowed diagnostics for the long runs (means per $\sim 1/6$ of training):

The mechanism is clear:

- **ratio** ≈ 1.0000 and **grad_norm** $\approx 5e-4$ **throughout**. With the tiny flow clip (**clip_range** $=\pm 10^{-4}$) and per-dimension-averaged Gaussian log-probs, each PPO step moves the policy a



Figure 2: Peak vs. final eval reward per config. The longer it trains, the more of the peak it gives back; short runs merely stopped near the top.

run	step	eval	train_rwd	expl_var	ratio	grad_norm
default-cfg	~226	0.837	0.815	-0.05	1.0000	8e-4
default-cfg	~1137	0.810	0.774	0.03	1.0000	6e-4
default-cfg	~2500	0.792	0.758	0.02	1.0000	5e-4
rewarmup-10-6	~465	0.811	0.778	-0.01	1.0000	5e-4
rewarmup-10-6	~5143	0.787	0.688	0.02	1.0000	6e-4

hair. *Nothing explodes*—so the collapse is not LR/clip instability but the slow accumulation of a *consistently biased* direction.

- **That biased direction is “maximize PickScore with no anchor.”** `kl_beta=0`, so nothing pulls the policy back toward the base model. This is the textbook RLHF over-optimization curve: proxy reward and even the eval metric rise, peak, then the policy walks off the manifold the reward model was trained on and the score decays. Eval peaks (step ~290–1113) *before* train reward fully turns over—a hallmark of reward hacking rather than learning.

Implication. Every knob swept so far (warmup length, re-warmup cadence, critic-follow) is downstream of the real problem. None addresses the missing anchor, so none stops the collapse—and `rewarmup-10-6` (the most aggressive re-warmup) ran longest and collapsed *hardest*.

4 Finding 3 — The critic barely earns its keep

- **Standalone critic ≈ 0 .** Because PickScore variance is tiny, `value_loss` looks “good” (~ 0.001) but that is an illusion: the critic just tracks the running *mean* reward (`value_mean  $\approx$  train_reward` in every window). So `advantage =  $R - V \approx R - \text{mean}(R)$` —PPO here is effectively **REINFORCE with a mean baseline**, buying nothing over a GRPO-style baseline.
- **Backbone critic (Scheme B) is the only critic with signal**—explained variance climbs to ~ 0.1 – 0.2 within 200 steps. This matches the Explicit-Critic-Guidance thesis (a state-aligned backbone value head is a genuinely better critic) and is the most promising structural lead, but every backbone run so far is ≤ 227 steps—*unproven past the peak*.

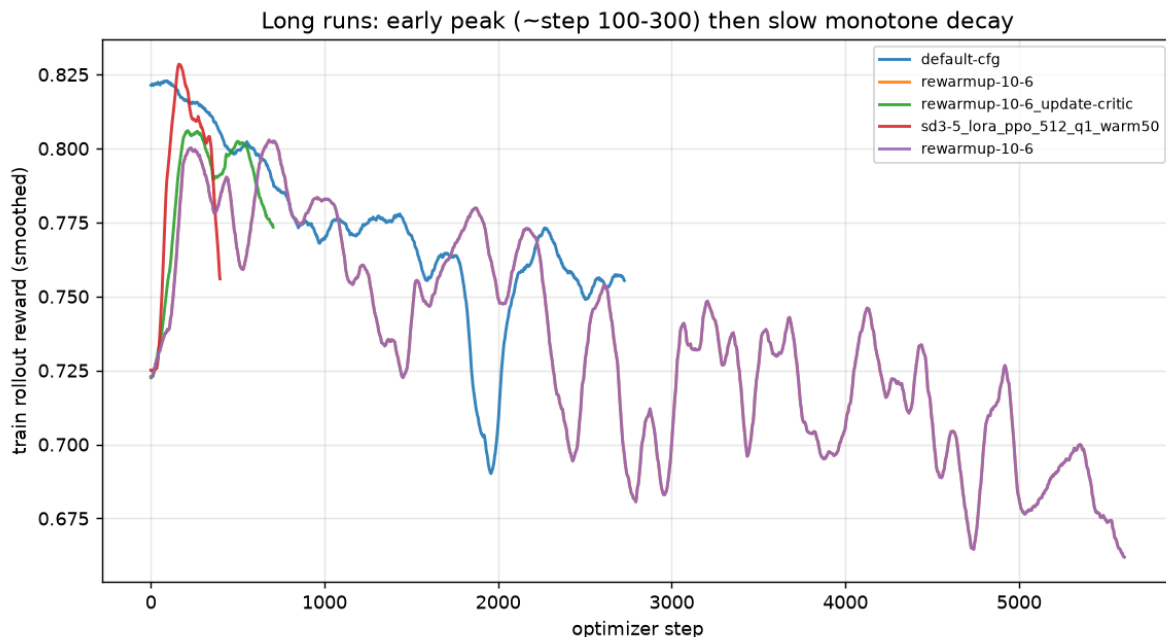


Figure 3: Smoothed train rollout reward for the long runs: an early peak (step $\sim 100-300$) then slow monotone decay over thousands of steps.

- **Single-SDE (bandit) critic is negative** (down to $-0.5 \dots -5$ early, settling slightly below 0). With one stochastic step the per-state value target is high-variance and the critic ends up *worse* than a constant. That run also had the weakest reward (train 0.78) and still collapsed. **Single-SDE is not the place to debug right now.**

5 Per-axis read

6 Recommendation — fix variables, solve collapse first

The core problem: ~ 6 axes are co-varied while the single most important one (`kl_beta`) is pinned at 0 and the success metric (sustained eval reward) is not what is being optimized. Change one thing per phase; hold the rest frozen.

Variables to FREEZE (the debug baseline).

- **Algorithm:** `sde_step_selection = global` (standard full-SDE PPO); drop single-SDE until a baseline works.
- **Warmup:** `critic_warmup_steps = 30`, **no periodic re-warmup** (`critic_warmup_interval = 0`), `update_critic_in_optimize = True`.
- **Critic structure:** pick ONE; recommended `standalone-512` as the cheap control for Phase 1 (so the anchor effect is isolated), backbone held for Phase 2.
- **Sampling:** `guidance_scale = 1.0`, `group_size = 1`, `num_inference_steps = 10`, current LRs.
- **Success metric:** **sustained eval/default/reward_pick_score_mean** over a long horizon (e.g. ≥ 1500 steps), *not* train reward and *not* the transient peak.

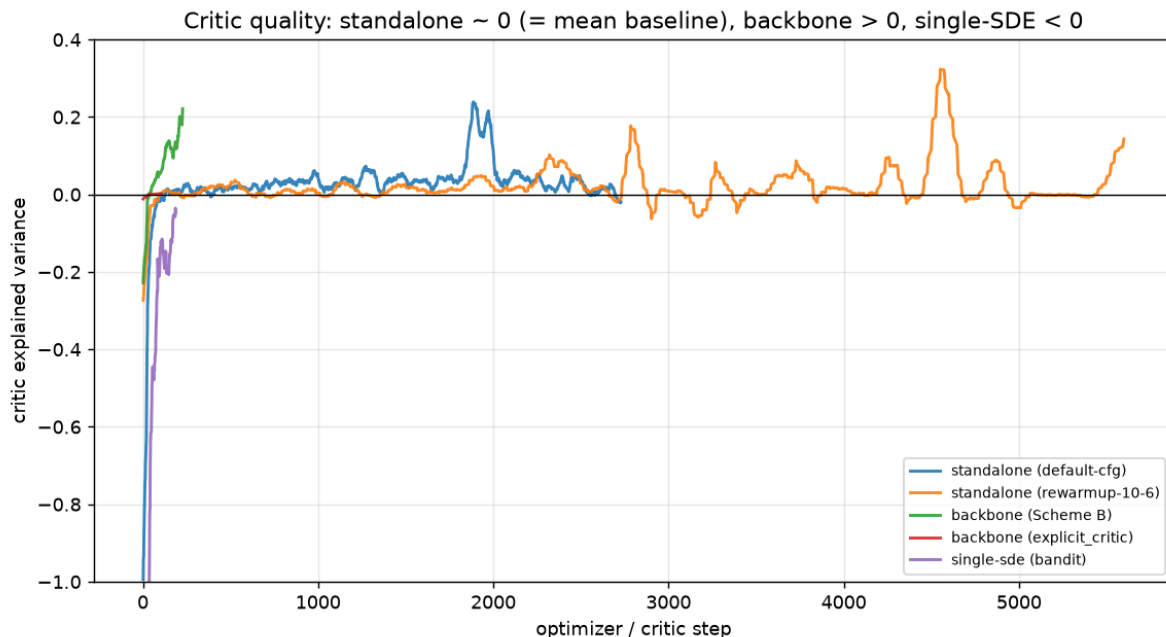


Figure 4: Critic explained variance (1 = perfect, 0 = no better than the mean, < 0 = worse than the mean). Standalone hovers at 0; backbone (Scheme B) climbs to ~ 0.2 ; single-SDE is negative.

Phase 1 (first): add the KL anchor. Sweep $\text{kl_beta} \in \{1\text{e-}3, 3\text{e-}3, 1\text{e-}2, 3\text{e-}2\}$ with $\text{kl_type} = \text{v-based}$ (cheaper; $\text{x-based} \sim 1\text{e-}2$ as a second arm), everything else at the frozen baseline. Expect a bias-variance band: too small $\Rightarrow$ still collapses; too large $\Rightarrow$ never rises. ~ 4 runs, one variable, directly targets the demonstrated failure—and has never been run.

Phase 2: make the critic real (only after Phase 1 holds). With the best kl_beta fixed, compare **standalone vs backbone (Scheme B)**, judged by explained variance (must be clearly > 0) and final eval reward. Optionally add the perturbed-state / branch- K critic target (see docs/ppo/ppo_analysis §“DiffAC-style perturbed-state critic”) to push explained variance higher.

Phase 3: revisit deferred knobs. Re-introduce re-warmup, updC=False alternation, single-SDE, or 40-step inference one at a time; keep only what beats the Phase-2 baseline on the sustained-eval metric.

Why this order. The collapse is *universal and anchor-shaped*, so no critic/warmup tuning can succeed until the anchor exists (Phase 1). Once the policy is prevented from walking off-manifold, critic quality (Phase 2) determines whether PPO actually beats GRPO; everything else is a refinement (Phase 3). Fix the algorithm to **global full-SDE PPO with a standalone critic** as the control, and vary **only kl_beta** first.

Reproduce

Histories were pulled from W&B project Flow-PPO into `.scratch/ppo_wandb/{runs_raw.json, collapse_stats.json, hist/*.csv}` with the repository `.venv` (wandb, matplotlib, pandas). Figures were regenerated from `.scratch/ppo_wandb/hist/*.csv` into `docs/ppo/figures/`; raw per-run data stays under `.scratch/` (git-ignored).

Table 3: What the data says about each knob.

Axis	Variants tried	Verdict from data
KL anchor kl_beta	only 0 (all runs)	Untested and almost certainly the root cause. Highest-priority variable.
Warmup steps	15, 30, 50	No effect on collapse; warm50 peaked the same and rolled over.
Periodic re-warmup	off, 10/5, 10/6, 100/10	Does <i>not</i> prevent collapse; rewarmup-10-6 ran longest and collapsed hardest. Complexity for no benefit.
Critic-follows-policy updC	True / False	updC=True decayed slightly more gently but still no growth; second-order.
Value-net structure	standalone / backbone / branches	Backbone > standalone on explained variance — the one structural win, but unproven long-term.
Algorithm	global / per_sample	Single-SDE underperforms (negative expl. var, weakest reward). Defer.
guidance_scale	1.0 / 4.5	4.5 reached the highest peak (0.838) but collapsed like the rest; not a fix.
infer steps	10 / 40	Backbone-40 too short to conclude.

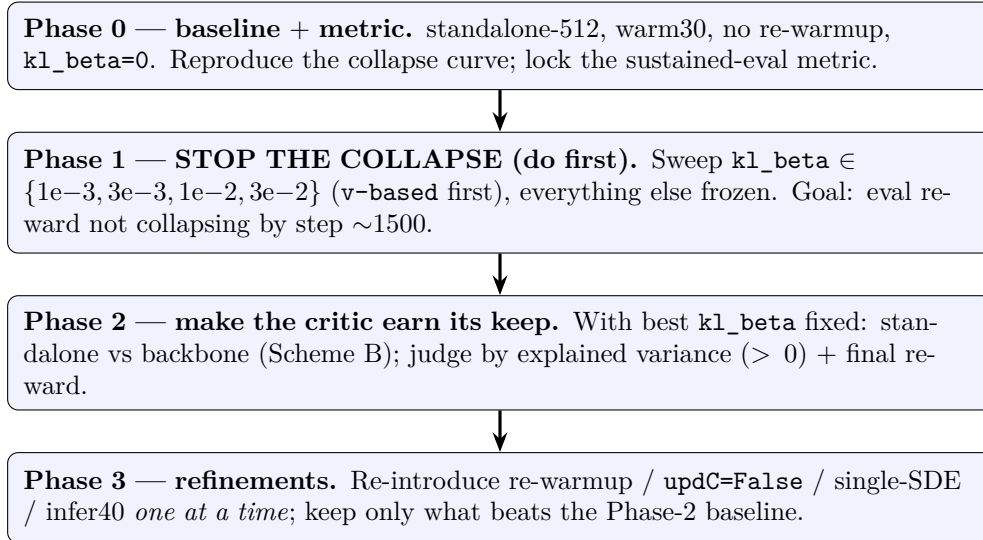


Figure 5: Phased plan: one variable per phase, collapse solved before critic tuning.